operating temperature as defined in the engineering specification. Exactly the same reading related to a different object—a cooling pump with a maximum operating temperature of 70° now becomes a critical alert, requiring immediate attention.

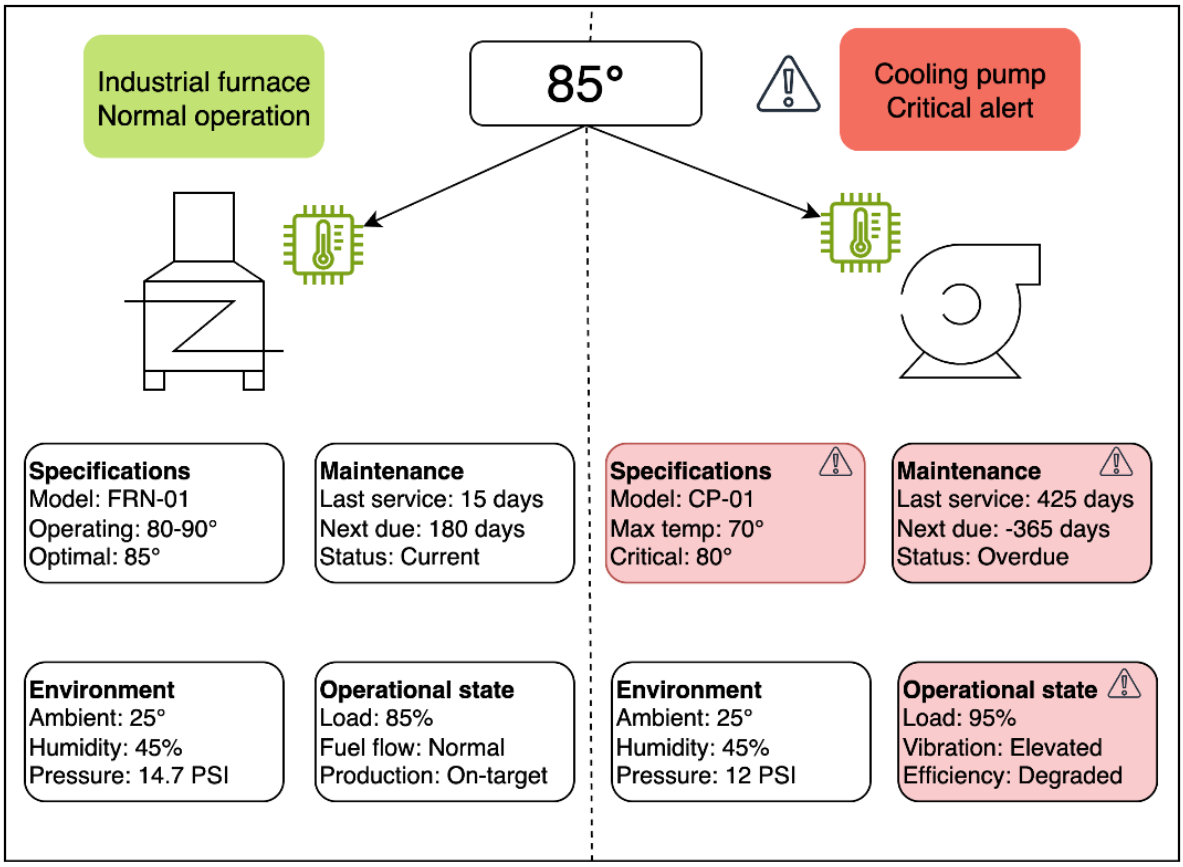


Figure 5.3 The importance of context in interpreting raw data and transforming it into information and knowledge.

A trained operator recognizes this distinction instantly, but their expertise masks significant complexity. Imagine it's your first day and you encounter this 85° pump reading. You must first locate the pump's operating specifications by logging into multiple systems, searching by model or serial number, or hunting through physical documentation. After determining the pump exceeds safe temperature, you need to understand system-wide impacts by consulting piping diagrams stored in a different system. To diagnose the root cause, you must retrieve maintenance history from yet another database —after figuring out how this pump is identified there. This fragmentation